

EXPERIMENT-29 Forward Chaining

Aim

To implement **Forward Chaining** in Prolog.

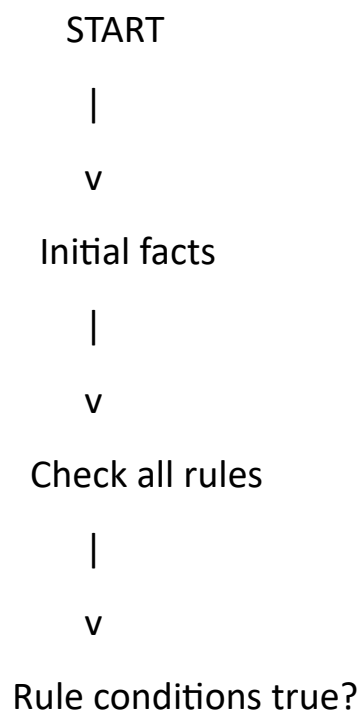
Objectives

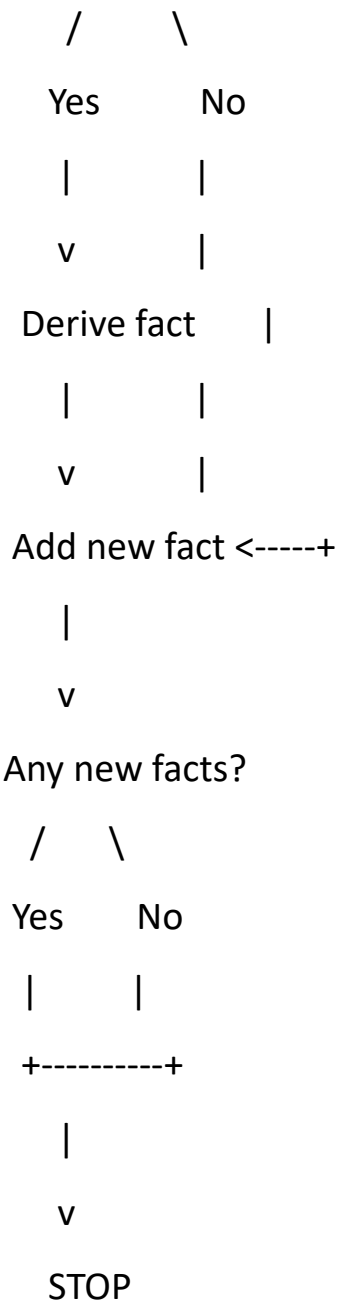
- To understand data-driven reasoning.
- To derive new facts from existing facts and rules.
- To demonstrate inference from known facts toward conclusions.

Algorithm

1. Store initial facts.
2. Check the rules whose conditions match the facts.
3. Derive new facts.
4. Add the newly derived facts to the knowledge base.
5. Continue until no new facts can be derived.
6. Check whether the required conclusion is present.

Flowchart





Program

% Forward Chaining

fact(fever).

fact(cough).

rule(flu, [fever, cough]).

```
rule(rest, [flu]).
```

```
forward :-
```

```
    forward_chain([fever, cough], Facts),  
    write('Derived facts: '),  
    write(Facts), nl.
```

```
forward_chain(Facts, FinalFacts) :-
```

```
    find_new_fact(Facts, NewFact),  
    \+ member(NewFact, Facts),  
    !,  
    forward_chain([NewFact | Facts], FinalFacts).
```

```
forward_chain(Facts, Facts).
```

```
find_new_fact(Facts, Conclusion) :-
```

```
    rule(Conclusion, Conditions),  
    all_present(Conditions, Facts).
```

```
all_present([], _).
```

```
all_present([H | T], Facts) :-
```

```
    member(H, Facts),  
    all_present(T, Facts).
```

Query

?- forward.

Output

```
HelloWorld.pl x
rule(rest, [flu]).

% Forward chaining
forward :-
    forward_chain([fever, cough], Facts),
    write('Derived facts: '),
    write(Facts),
    nl.

forward_chain(Facts, FinalFacts) :-
    find_new_fact(Facts, NewFact),
    \+ member(NewFact, Facts),
    !,
    forward_chain([NewFact|Facts], FinalFacts).

forward_chain(Facts, Facts).

% Check whether all conditions are present
find_new_fact(Facts, Conclusion) :-
    rule(Conclusion, Conditions),
    all_present(Conditions, Facts).

all_present([], _).

all_present([H|T], Facts) :-
    member(H, Facts),
    all_present(T, Facts).
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

Derived facts: [rest,flu,fever,cough]

Result

Forward chaining was successfully implemented by deriving new facts from known facts.

Conclusion

Forward chaining is a data-driven reasoning technique where conclusions are derived progressively from available facts.